

Advanced Optimization

Chapter 12 — Theory of Constrained Optimization

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 12

The problem, and the vocabulary that carries the rest of the book

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad c_i(x) = 0, \quad i \in \mathcal{E}; \quad c_i(x) \geq 0, \quad i \in \mathcal{I}; \quad \Omega = \{x : \text{feasible}\}. \quad (12.1)$$

- ▶ **Local / strict / isolated solutions** as in Ch. 2, intersected with Ω . Constraints can *create* many isolated local solutions out of a smooth f (Figure 12.1).
- ▶ **Active set** (Definition 12.1): $\mathcal{A}(x) = \mathcal{E} \cup \{i \in \mathcal{I} : c_i(x) = 0\}$. The single most-used object of Chapters 12–19.

Smoothness: the boundary has kinks; the functions need not

The diamond $\|x\|_1 \leq 1$ has four corners — but is described by four *linear* constraints (12.6). Conversely, nonsmooth objectives become smooth constrained problems: $\min \max(x^2, x) \rightarrow \min t \text{ s.t. } t \geq x, t \geq x^2$ (12.8). This trick carries ℓ_1/ℓ_∞ fitting into Ch. 17.

Where this sits relative to Boyd

You know convex KKT: with Slater, necessary *and* sufficient, zero duality gap. This chapter rebuilds the theory *without convexity*: KKT becomes necessary only, it needs a **constraint qualification** to hold at all, and second-order conditions take over the sufficiency role. Watching what survives and what breaks is the point of the week.

One equality constraint: where $\nabla f = \lambda \nabla c$ comes from

Example 12.1: $\min x_1 + x_2$ s.t. $x_1^2 + x_2^2 - 2 = 0$. Solution $(-1, -1)^T$.

First-order reasoning: a step s keeps feasibility (to first order) iff $\nabla c_1(x)^T s = 0$ (12.12), and improves iff $\nabla f(x)^T s < 0$ (12.13). **No such direction exists** exactly when ∇f has no component along the constraint surface:

$$\nabla f(x^*) = \lambda_1^* \nabla c_1(x^*) \quad (\text{here } \lambda_1^* = -\tfrac{1}{2}). \quad (12.10)$$

Equivalently $\nabla_x \mathcal{L}(x^*, \lambda_1^*) = 0$ for $\mathcal{L} = f - \lambda_1 c_1$.

Two sobering observations, both from this tiny example

- ▶ (12.10) also holds at $(1, 1)^T$ — the *maximizer*. First-order conditions are **necessary, never sufficient**, exactly as in Ch. 2.
- ▶ Rewrite the constraint as $2 - x_1^2 - x_2^2 = 0$: same problem, but λ_1^* flips from $-\frac{1}{2}$ to $+\frac{1}{2}$. **For equality constraints the multiplier's sign is an artifact of the algebra** — no sign condition can be imposed.

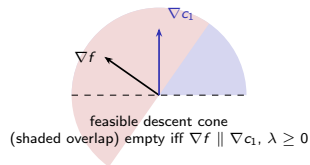
One inequality: the sign appears; complementarity unifies

Example 12.2: same objective, constraint $2 - x_1^2 - x_2^2 \geq 0$ (disk). Feasibility to first order: $c_1(x) + \nabla c_1(x)^T s \geq 0$ (12.19).

Case I (x interior, $c_1 > 0$): every small s is feasible $\Rightarrow$ no descent direction iff $\nabla f(x) = 0$. Unconstrained optimality reappears.

Case II (x on the boundary, $c_1 = 0$): need $\nabla f^T s < 0$ (open half-space) and $\nabla c_1^T s \geq 0$ (closed half-space). Intersection empty iff the two gradients point the same way:

$$\nabla f(x) = \lambda_1 \nabla c_1(x), \quad \boxed{\lambda_1 \geq 0}. \quad (12.21)$$



A negative λ_1 would leave an entire open half-plane of feasible descent directions.

Both cases in one line: complementarity

$$\nabla_x \mathcal{L}(x^*, \lambda_1^*) = 0, \quad \lambda_1^* \geq 0, \quad \lambda_1^* c_1(x^*) = 0 \quad (12.22-12.23)$$

Interior: $c_1 > 0$ forces $\lambda_1^* = 0$ (and the condition collapses to $\nabla f = 0$). Boundary: λ_1^* may be positive. The multiplier switches the constraint on and off.

Two inequalities: the picture generalizes, multiplier signs diagnose optimality

Example 12.3: add $x_2 \geq 0$ to the disk problem; solution $x^* = (-\sqrt{2}, 0)^T$, both constraints active.

- ▶ Feasible descent requires $\nabla c_i(x)^T d \geq 0$ for both active i and $\nabla f^T d < 0$ (12.25). At x^* the first two force d into the quadrant spanned by $\nabla c_1, \nabla c_2$ — where $\nabla f^T d \geq 0$ always. No escape: $\nabla f(x^*) = \lambda_1^* \nabla c_1 + \lambda_2^* \nabla c_2$ with $\lambda^* = (\frac{1}{2\sqrt{2}}, 1) \geq 0$.
- ▶ At the *non*-solution $(\sqrt{2}, 0)^T$ (also both active): $\nabla_x \mathcal{L} = 0$ is solvable only with $\lambda_1 = -\frac{1}{2\sqrt{2}} < 0$ — **the negative multiplier is the certificate of non-optimality**, and indeed $d = (-1, 0)^T$ is a feasible descent direction.
- ▶ At $(1, 0)^T$ (only c_2 active): the inactive disk constraint contributes nothing; complementarity zeroes its multiplier.

What the examples bought us

A complete guess at the general conditions: stationarity of $\mathcal{L}$, feasibility, $\lambda \geq 0$ on inequalities, complementarity. What is missing is *rigor about the linearization* — when do first-order feasibility conditions actually describe the feasible set? That is §12.2.

Two cones: one geometric, one algebraic

Tangent cone $T_{\Omega}(x)$ (Definition 12.2) — geometry only

d is a tangent if some feasible sequence $z_k \rightarrow x$ and scalars $t_k \downarrow 0$ give $(z_k - x)/t_k \rightarrow d$.
Depends on the set Ω , not on how it is written.

Linearized feasible directions $\mathcal{F}(x)$ (Definition 12.3) — algebra

$$\mathcal{F}(x) = \{d : \nabla c_i(x)^T d = 0, i \in \mathcal{E}; \quad \nabla c_i(x)^T d \geq 0, i \in \mathcal{A}(x) \cap \mathcal{I}\}$$

Depends on the *functions* c_i chosen to describe Ω .

T_{Ω} is intrinsic; $\mathcal{F}$ is what algorithms can see. Optimality theory works only where the two agree — and they need not.

When the linearization lies, and the qualification that forbids it

Same set, different descriptions, different $\mathcal{F}$

- ▶ Circle at $(-\sqrt{2}, 0)$: standard description gives $\mathcal{F} = T_\Omega =$ the vertical axis. Describe the *same circle* by $(x_1^2 + x_2^2 - 2)^2 = 0$: now $\nabla c \equiv 0$ on the circle, so $\mathcal{F} = \mathbb{R}^2 \supsetneq T_\Omega$.
- ▶ Circle intersected with its tangent line = a single point (12.32): $T_\Omega = \{0\}$, yet $\mathcal{F} =$ a whole line.

In both cases the linearization *lies* about the geometry — and any optimality condition built from gradients would inherit the lie.

Constraint qualification = “the linearization tells the truth”

LICQ (Definition 12.4): the active gradients $\{\nabla c_i(x), i \in \mathcal{A}(x)\}$ are linearly independent. Both pathologies violate it (a zero gradient, or duplicated directions).

Other qualifications (§12.6, keywords): purely *linear* constraints need none at all (Lemma 12.7 — why LP and QP theory is cleaner); MFCQ is weaker than LICQ and still suffices; under MFCQ multipliers exist but may form a set rather than a point.

Theorem 12.1: the KKT conditions

Lagrangian $\mathcal{L}(x, \lambda) = f(x) - \sum_{i \in \mathcal{E} \cup \mathcal{I}} \lambda_i c_i(x)$ (12.33).

Theorem 12.1 (First-Order Necessary Conditions)

If x^* is a local solution, $f, c_i \in C^1$, and **LICQ holds at x^*** , then there is λ^* with

(a) $\nabla_x \mathcal{L}(x^*, \lambda^*) = 0$	stationarity	
(b) $c_i(x^*) = 0, i \in \mathcal{E};$	(c) $c_i(x^*) \geq 0, i \in \mathcal{I}$	feasibility
(d) $\lambda_i^* \geq 0, i \in \mathcal{I}$		dual feasibility
(e) $\lambda_i^* c_i(x^*) = 0, i \in \mathcal{E} \cup \mathcal{I}$		complementarity

(12.34)

- ▶ By (e), only active constraints appear in (a): $\nabla f(x^*) = \sum_{i \in \mathcal{A}(x^*)} \lambda_i^* \nabla c_i(x^*)$ (12.35).
- ▶ **Strict complementarity** (Definition 12.5): $\lambda_i^* > 0$ for every active inequality — when it holds, algorithms identify $\mathcal{A}(x^*)$ cleanly and converge faster.
- ▶ Under LICQ the multiplier λ^* is *unique*.
- ▶ Boyd's world: convex + Slater $\Rightarrow$ KKT is also *sufficient* and CQ comes cheap. Here: necessary only; sufficiency waits for second order (§12.5).

The proof is three pillars and one assembly step

- 1 Lemma 12.2:** always $T_{\Omega}(x^*) \subset \mathcal{F}(x^*)$; under LICQ, $T_{\Omega}(x^*) = \mathcal{F}(x^*)$.
(Easy half: Taylor along a feasible sequence. Hard half: LICQ + implicit function theorem to *build* a feasible curve realizing each $d \in \mathcal{F}$.)
- 2 Theorem 12.3 (no CQ needed):** x^* local solution $\Rightarrow \nabla f(x^*)^T d \geq 0$ for all $d \in T_{\Omega}(x^*)$.
(A descending tangent direction would generate a feasible sequence with $f(z_k) < f(x^*)$.)
- 3 Farkas' Lemma 12.4:** for the cone $K = \{By + Cw : y \geq 0\}$ and any g : either $g \in K$, or there is d with $g^T d < 0$, $B^T d \geq 0$, $C^T d = 0$ — **never both**.

Assembly

LICQ: $T_{\Omega} = \mathcal{F}$ (pillar 1). Optimality: $\nabla f^T d \geq 0$ on $T_{\Omega} = \mathcal{F}$ (pillar 2). So *no* d separates ∇f from the active-gradient cone — Farkas (pillar 3) then forces

$$\nabla f(x^*) \in K = \left\{ \sum_{i \in \mathcal{A} \cap \mathcal{I}} \lambda_i \nabla c_i + \sum_{i \in \mathcal{E}} \lambda_i \nabla c_i : \lambda_i \geq 0 \text{ on } \mathcal{I} \right\}$$

— which is (12.34a,d); feasibility and complementarity come from the construction. $\square$

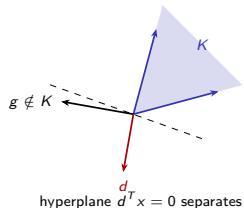
Farkas' lemma: the separating hyperplane doing the real work

Lemma 12.4

$K = \{By + Cw : y \geq 0\}$ (conic hull of inequality columns + span of equality columns). For any g : **either** $g \in K$ **or** $\exists d$: $g^T d < 0$, $B^T d \geq 0$, $C^T d = 0$. Not both.

Proof sketch. Both: impossible by direct computation ($0 > d^T g = (B^T d)^T y + (C^T d)^T w \geq 0$). If $g \notin K$: project g onto the *closed* cone K ; the closest point $\hat{s}$ satisfies $\hat{s}^T(\hat{s} - g) = 0$, and $d = \hat{s} - g$ separates. Closedness of K is the one nontrivial fact.

Boyd connection: this is the same separating-hyperplane machinery that proves strong duality — here deployed for *existence of multipliers*.



Why the proof matters more than usual

Every chapter from here on trades in multipliers: QP (Ch. 16), SQP (Ch. 18), interior point (Ch. 19) all *solve* the KKT system. Knowing that λ^* exists because of a projection argument — not magic — is what makes the later linear algebra meaningful.

Second order: tiebreaking on the critical cone

KKT leaves “undecided” directions: $w \in \mathcal{F}(x^*)$ with $w^T \nabla f(x^*) = 0$ — first order says nothing there. Collect them in the **critical cone**:

$$\mathcal{C}(x^*, \lambda^*) : \quad \nabla c_i^T w = 0 \quad (i \in \mathcal{E}, \text{ and } i \in \mathcal{A} \cap \mathcal{I} \text{ with } \lambda_i^* > 0); \quad \nabla c_i^T w \geq 0 \quad (i \in \mathcal{A} \cap \mathcal{I}, \lambda_i^* = 0). \quad (12.53)$$

Indeed $w \in \mathcal{C} \Rightarrow w^T \nabla f(x^*) = 0$ (12.55): critical directions are exactly the ones that cling to the strongly active constraints.

Theorem 12.5 (necessary) and Theorem 12.6 (sufficient)

$$(N): w^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) w \geq 0 \quad \forall w \in \mathcal{C}; \quad (S): w^T \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) w > 0 \quad \forall w \in \mathcal{C}, w \neq 0.$$

(N) needs LICQ; (S) needs no constraint qualification at all and delivers a *strict* local solution with a quadratic growth estimate $f(z) \geq f(x^*) + \frac{\sigma}{4} \|z - x^*\|^2$ nearby.

- ▶ Curvature of the **Lagrangian**, not of f — the constraints’ curvature enters through $-\sum \lambda_i^* \nabla^2 c_i$.
- ▶ Only on $\mathcal{C}$, not on all of $\mathbb{R}^n$ — an indefinite $\nabla_{xx}^2 \mathcal{L}$ is fine if its negative directions are infeasible.
- ▶ Gap between (N) and (S): the usual $\geq$ vs. $>$, same as Ch. 2.

A worked check, and the projected-Hessian form

Example (12.72): $\min -0.1(x_1 - 4)^2 + x_2^2$ s.t. $1 - x_1^2 - x_2^2 \geq 0$. At $x^* = (1, 0)^T$: KKT holds with $\lambda_1^* = 0.3$,

$$\nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) = \begin{bmatrix} -0.4 & 0 \\ 0 & 1.4 \end{bmatrix} \text{ (indefinite!)}, \quad \mathcal{C} = \{(0, w_2)^T\} \Rightarrow w^T \nabla_{xx}^2 \mathcal{L} w = 1.4 w_2^2 > 0.$$

Theorem 12.6: strict local solution. **The cone does the work the Hessian cannot** — the negative-curvature direction $(1, 0)$ points out of the feasible set.

Projected Hessians (when strict complementarity holds)

Then $\mathcal{C} = \text{Null } A(x^*)$, A = matrix of active gradients. With Z spanning that null space (e.g. from QR of A^T), the conditions become plain matrix statements:

$$(N): Z^T \nabla_{xx}^2 \mathcal{L} Z \succeq 0; \quad (S): Z^T \nabla_{xx}^2 \mathcal{L} Z \succ 0.$$

The *reduced Hessian* — checkable by a Cholesky attempt, and the exact object that Chapters 16 and 18 factor inside their algorithms.

The normal cone: KKT with the algebra stripped away

Definition 12.7 and Theorem 12.8

$N_{\Omega}(x) = \{v : v^T w \leq 0 \ \forall w \in T_{\Omega}(x)\}$ — everything at $\geq 90^\circ$ to all tangents. If x^* is a local minimizer:

$$-\nabla f(x^*) \in N_{\Omega}(x^*). \quad (12.78)$$

Proof: two lines from Theorem 12.3 and the definition.

- ▶ Purely geometric — no c_i , no CQ, no multipliers. (Boyd readers: this is the optimality condition over a convex set, now stated for any closed set.)
- ▶ **Lemma 12.9:** under LICQ, $N_{\Omega}(x^*) = -\{\sum_{i \in \mathcal{A}} \lambda_i \nabla c_i : \lambda_i \geq 0 \text{ on } \mathcal{I}\}$ — the geometric statement and the KKT statement are *the same statement*, translated by Farkas.
- ▶ Mental model for everything that follows: **at a solution, $-\nabla f$ is absorbed by the normal cone; multipliers are its coordinates in the active-gradient basis.**

Sensitivity: what a multiplier is worth

Perturb an active constraint: $c_i(x) \geq -\epsilon \|\nabla c_i(x^*)\|$. Tracking the solution family $x^*(\epsilon)$ through the KKT conditions gives

$$\frac{d f(x^*(\epsilon))}{d\epsilon} = -\lambda_i^* \|\nabla c_i(x^*)\|. \quad (12.80)$$

- ▶ λ_i^* is the **price of the constraint**: how much objective one unit of relaxation buys. Inactive $\Rightarrow$ price 0 (complementarity, again).
- ▶ **Definition 12.8**: active with $\lambda_i^* > 0$: *strongly active (binding)*; active with $\lambda_i^* = 0$: *weakly active* — removing it changes nothing to first order.
- ▶ The estimate needs strict complementarity + second-order conditions to be made rigorous — the honest fine print.

Why engineers already believe this

Shadow prices are bread and butter: in electricity markets, the *locational marginal price is the Lagrange multiplier of the nodal power-balance constraint* of an OPF. When you quote an LMP, you are quoting λ_i^* of (12.34). Chapter 12 is where that number gets its meaning.

Duality in the nonconvex world (§12.9)

For $\min f$ s.t. $c(x) \geq 0$: dual function and dual problem

$$q(\lambda) = \inf_x \mathcal{L}(x, \lambda), \quad \max_{\lambda \geq 0} q(\lambda). \quad (12.82, 12.84)$$

- ▶ **Theorem 12.10:** q is *always* concave (infimum of affine functions of λ) — convexity of the dual costs nothing, even for wildly nonconvex primals.
- ▶ **Theorem 12.11 (weak duality):** $q(\bar{\lambda}) \leq f(\bar{x})$ for any feasible pair. Three lines.
- ▶ **Theorem 12.12 (Wolfe):** if the problem *is* convex and $(\bar{x}, \bar{\lambda})$ satisfies KKT, then $\bar{\lambda}$ solves the dual and $q(\bar{\lambda}) = f(\bar{x})$ — Boyd's strong duality recovered. Theorem 12.13 gives a partial converse.
- ▶ Worked micro-example (12.85): $\min \frac{1}{2}(x_1^2 + x_2^2)$ s.t. $x_1 \geq 1 \Rightarrow q(\lambda) = -\frac{1}{2}\lambda^2 + \lambda$, dual solution $\lambda = 1$ — matching the primal multiplier.

The nonconvex caveat

Without convexity, $\max q$ can sit strictly below $\min f$: a **duality gap**. The dual is still useful — bounds, relaxations, decomposition — but it certifies rather than solves. (This gap is precisely why “solving the SDP/SOCP relaxation of AC-OPF” and “solving AC-OPF” are different claims.)

Chapter 12 in one table

Condition	Needs	Delivers	Where
$\nabla f^T d \geq 0$ on T_Ω	nothing	raw necessity	Thm 12.3
$T_\Omega = \mathcal{F}$	LICQ (or linear c_i)	linearization truthful	Lem 12.2, 12.7
KKT (12.34)	LICQ	multipliers exist, unique	Thm 12.1
$w^T \nabla^2 \mathcal{L} w \geq 0$ on $\mathcal{C}$	LICQ	2nd-order necessity	Thm 12.5
$w^T \nabla^2 \mathcal{L} w > 0$ on $\mathcal{C}$	no CQ	strict local solution	Thm 12.6
$-\nabla f \in N_\Omega$	nothing	geometric necessity	Thm 12.8
$df/d\epsilon = -\lambda_i^* \ \nabla c_i\ $	strict compl. + 2nd order	shadow prices	(12.80)
q concave; $q \leq f$	nothing	bounds always	Thm 12.10–11
zero gap	convexity	Boyd's world	Thm 12.12–13

Three ideas to carry forward

A constraint qualification is a promise that the linearization tells the truth; KKT is what Farkas extracts from that promise. Second-order conditions test the Lagrangian's curvature only on the critical cone — indefiniteness elsewhere is free. A multiplier is a price; complementarity says only scarce goods have prices.

Homework

Derivations

- 1 In Example 12.1, rewrite the constraint as $2 - x_1^2 - x_2^2 = 0$ and verify that λ_1^* flips sign while the solution is unchanged.
- 2 Show that $T_\Omega(x)$ and $\mathcal{F}(x)$ are cones, and prove Lemma 12.2(i) ($T_\Omega \subset \mathcal{F}$; Taylor along the sequence).
- 3 Prove the “not both” half of Farkas’ lemma, and complete the projection half using $\hat{s}^T(\hat{s} - g) = 0$ (12.48).
- 4 Work Example 12.6 (the diamond, (12.36)): find the solution, active set, and multipliers; check LICQ.
- 5 Verify all claims of Example (12.72): KKT at $(1, 0)$, $\mathcal{C}$, and the second-order sufficient conditions; then explain in one sentence why the indefiniteness of $\nabla_{xx}^2 \mathcal{L}$ is harmless.
- 6 Prove Theorem 12.10 (concavity of q) and Theorem 12.11 (weak duality), and rederive the dual of (12.85) by hand.

Numerical

7. Solve the diamond problem (12.36) with an off-the-shelf solver; extract the reported multipliers; verify (12.34) to machine precision and match your item-4 answer.
8. *Sensitivity check.* Re-solve (12.72) with the constraint relaxed to $1 + \epsilon - x_1^2 - x_2^2 \geq 0$, $\epsilon \in \{10^{-1}, \dots, 10^{-4}\}$; confirm the slope of $f(x^*(\epsilon))$ against the prediction of (12.80).

Lecture track continues with Ch. 16 (quadratic programming). Next deck: Ch. 13 — *linear programming (self-study)*.